

VentureMiner AI

RAG Architecture

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Document 10 — RAG Architecture

Retrieval-Augmented Generation (RAG) is the only path by which an agent can ground a claim in external knowledge. This document defines the corpus, the index, the retrieval pipeline, the citation model, and the evaluation framework.

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1. Purpose & Scope

This document is the RAG architectural contract. It governs the corpus, the index, the retrieval, and the citation discipline that all agents depend on.

2. Why RAG (and only RAG)

- Grounded claims. The model cannot invent a number if it must cite a chunk.
- Freshness. Sources can be re-fetched without retraining.
- Tenant isolation. Each workspace has its own corpus slice.
- Auditability. A citation is a link to a chunk that links to a source URL.

We forbid direct LLM knowledge claims for any quantitative or time-sensitive fact. Anything not in the corpus must be marked as "inferred" and reviewed by the verifier.

3. Corpus

3.1 Composition

- Public web — fetched on demand for known sources; cached in the corpus.
- Source API — Reddit, X, GitHub, AppStores, G2, HN, etc. — pushed via source connectors.
- Internal documents — workspace-uploaded PDFs, docs, sheets.
- Knowledge packs — curated sets (industry reports, market data) per vertical.

3.2 Lifecycle

- Capture: source connector fetches → write to corpus.raw.
- Normalize: clean, dedupe, extract text + metadata.
- Chunk: split into retrieval units (Section 5).
- Embed: generate embeddings (Section 6).
- Index: upsert to vector + lexical indexes (Section 7).
- Serve: retrieve on demand (Section 8).

4. Ingestion pipeline

```

flowchart LR
  Src[Source] --> Conn[Connector]
  Conn --> Raw[(corpus.raw)]
  Raw --> Norm[Normalize]
  Norm --> Chunk[Chunk]
  Chunk --> Emb[Embed]
  Emb --> VS[(vector index)]
  Emb --> LS[(lexical index)]
  Conn --> Meta[Metadata DB]
  Meta --> VS
  Meta --> LS

```

- Backpressure: if embedding queue depth > N, pause ingest; surface backpressure metric.
- Replay: every step is idempotent; ingestion is replayable from corpus.raw.

5. Chunking

- Default chunker: recursive character with overlap.
- Chunk size: 800 tokens target, 1200 max, 200 min.
- Overlap: 200 tokens.
- Structure-aware chunker for HTML, Markdown, and PDF (preserves headings).
- Code-aware chunker for code (function boundaries).
- Tables: preserved as Markdown tables within a chunk.

- v2 candidate: a self-hosted BGE-M3 for cost.
- Normalization: L2-normalized; cosine similarity.
- Metadata: source URL, fetch timestamp, freshness class, language, document type.

7. Index

7.1 Vector index

- Storage: pgvector (v1), Qdrant (v2).
- Index type: HNSW (m=16, ef_construction=64), recast to ivfflat if rows < 100k.
- Sharding: by tenant (workspace).

7.2 Lexical index

- Engine: OpenSearch 2.x with BM25.
- Sharding: by tenant.

7.3 Hybrid retrieval

- Combine vector + lexical scores using reciprocal rank fusion (RRF, k=60).
- Boost documents by freshness and by source weight (configurable per source).

8. Retrieval pipeline

```
query → embed → ANN (vector) + BM25 (lexical) → RRF fuse → top 50 → rerank → top 10 → return
```

- Top-10 default; configurable per agent.
- Query rewriting for multi-turn: include prior turn context into a single query string.

9. Re-ranking

- Reranker model: bge-reranker-v2-m3 or Cohere Rerank 3.
- Goal: push the most relevant chunk to position 1.
- Cost: reranker calls are the second-largest variable cost after LLM synthesis.

10. Citation model

Every claim produced by an agent is bound to:

- A chunk_id (which contains a source_url and fetched_at).
- A confidence (high/med/low).
- A freshness_class (live, recent, stale, unknown).

The agent's output must include, for every claim:

```
{ "claim": "...", "citation": { "chunk_id": "...", "source_url": "...",
"fetch_at": "...", "confidence": "high" } }
```

The verifier audits the citation; ungrounded claims are rejected.

11. Freshness

- Live: fetched within the last 24h.
- Recent: 1–7d.
- Stale: 7–30d.
- Unknown: > 30d or no timestamp.

Agents MUST mark time-sensitive claims with a freshness class. The report surfaces freshness per section.

12. Quality evaluation

12.1 Metrics

- Recall@10 — does the chunk containing the answer appear in the top 10?
- Citation precision — what fraction of cited chunks are actually relevant?
- Citation recall — what fraction of claims have a citation?
- Freshness accuracy — does the marked freshness match the source's actual age?

12.2 Eval set

- 500 hand-labeled queries across verticals, refreshed quarterly.
- Regressions > 2% block release.

13. Per-tenant corpus

- Each workspace has its own collection (RC-).
- Tenant isolation enforced at index time (shard key) and at query time (filter).
- Cross-tenant retrieval is impossible by design.

14. Cost & scale

Embedding	\$80
Vector query	\$0.10 (pgvector on db.r6g.2xlarge)
Reranker call	\$20
Index build	\$5 / M chunks

Targets:

- 100M chunks in corpus at Year 1.
- p95 retrieval latency < 300ms.

Embedding provider down	Fallback to second provider; if both down, fail RAG call and surface "no evidence".
pgvector index corrupt	Rebuild from corpus.chunk (idempotent).
OpenSearch unhealthy	Degrade lexical; serve vector only.
Reranker down	Skip rerank; cap top-k lower.
Stale corpus	Schedule re-fetch; freshness flag updates.

16. Appendix

16.1 Glossary

Chunk	A retrieval unit (target 800 tokens)
RRF	Reciprocal rank fusion — combined ranking across indexes
Freshness	How recent the source is at retrieval time
Citation precision	Fraction of citations that point to a relevant chunk

16.2 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

16.3 Cross-references

- Multi-Agent: Document 08.
- Agent Specs: Document 09.
- Memory: Document 11.
- MCP: Document 12.
- Plugin Architecture: Document 13.

End of Document 10 — RAG Architecture. The citation discipline here is the foundation of every report and every score.